ChatGPT

NHS Data Integration & Reporting Standards for Digital Therapy Tools

Third-party **digital therapy tools** (like OTOS) must adhere to NHS integration standards and data governance requirements in order to be used within health and justice settings. Below we outline the key integration points (with NHS clinical systems and the NHS App), relevant data-security frameworks, and reporting standards. We also recommend how OTOS can structure dashboards and reports to meet **NHS England and Probation Service KPIs** (e.g. engagement, outcomes, safeguarding).

Integration with NHS Systems (APIs & Interoperability)

- **Integration with GP Clinical Systems:** NHS primary care systems such as **EMIS Web** and **TPP SystmOne** offer integration interfaces for approved third-party apps. Through NHS England's **IM1 API program**, a digital tool can **read patient information, extract data in bulk, and write data back to GP records** [1]. Each IM1 interface corresponds to one of the principal GP system suppliers (EMIS or SystmOne) [2]. For example, an approved app can search and retrieve a patient's medical record or add a consultation note in real-time via these APIs [3] [4]. To gain access, OTOS would need to undergo the **IM1 Pairing onboarding process**, including completing clinical and information governance prerequisites and a Solution Conformance Assessment List (SCAL) to ensure compatibility and data security [5]. Once the product passes this assurance and signs the interface licence, it can connect to live GP systems [6].

- **Standards and Open APIs (FHIR):** The NHS has embraced **FHIR (Fast Healthcare Interoperability Resources)** as the core standard for health data exchange. FHIR is an open global standard that makes it easier to **share healthcare data between systems** and ensures different applications can communicate patient data securely [7]. New NHS APIs (such as **GP Connect** for sharing GP records or appointment info) are built on FHIR, so any third-party tool should be designed to **consume and produce FHIR-formatted data** for future-proof integration [8] [9]. Using FHIR also aligns with NHS interoperability requirements in the DTAC (Digital Technology Assessment Criteria).

- **NHS Login and NHS App Integration:** To reach patients via the official **NHS App**, third-party services must integrate using NHS Digital's guidelines. This typically involves using **NHS Login** (the NHS's secure OAuth2/OpenID Connect identity system) for user authentication, and embedding the service within the NHS App's interface. For instance, the eConsult platform is available inside the NHS App, allowing users to access eConsult without leaving the NHS App; this **integration uses a standard NHS FHIR API for consultations** (the NHS Digital CDS FHIR specification) with eConsult acting as the API provider [10]. In practice, OTOS would need to undergo an NHS App partner onboarding, ensure **seamless single sign-on via NHS Login** [11], and meet technical requirements so that data (e.g. therapy session outcomes or messages) can be exchanged securely with NHS systems. Many digital health services are in the pipeline to integrate directly into the NHS App [12] [13], which underscores the importance of conforming to NHS integration standards and using approved APIs.

- **Other Integration Mechanisms:** In cases where direct real-time API write-back is not available, tools often use NHS integration services like **MESH (Message Exchange for Social Care and Health)** to send documents or data extracts to NHS systems. For example, eConsult uses MESH to deliver reports into GP practice workflows (e.g. sending a consultation summary PDF into the patient's record in EMIS or SystmOne) [14]. OTOS should be prepared to support such mechanisms for data transfer (e.g. sending outcome reports or alerts to clinicians) if required. Additionally, being part of the EMIS Partner Program or equivalent (as eConsult did) allows deeper integration such as writing back notes into the GP record instantly [15]. In summary, **compliance with NHS interoperability standards** – using FHIR where possible, and working within frameworks like GP Connect, IM1, NHS Login, and MESH – will be crucial for OTOS to integrate with NHS infrastructure.

Data Security & Compliance Frameworks

Any digital health tool handling NHS patient data or working with justice services must meet stringent **information governance (IG) and security standards**:

- **Data Security and Protection Toolkit (DSPT):** Formerly known as the IG Toolkit, the DSP Toolkit is an **online self-assessment** that all organizations with access to NHS patient data must complete annually [16]. It measures compliance against the National Data Guardian's 10 security standards [17]. In practice, OTOS will need to achieve a satisfactory DSPT score, demonstrating that it has robust data protection policies, access controls, training, and incident response in place. Completion of the DSP Toolkit is **mandatory to work with the NHS** or receive NHS patient data [18]. This ensures OTOS handles personal health information properly and gives NHS partners confidence in its data governance.

- **ISO 27001 Certification: ISO/IEC 27001** is the internationally recognized standard for Information Security Management Systems. Achieving ISO 27001 certification is highly recommended (and often expected) for digital health suppliers. It demonstrates that OTOS follows best-practice security controls and risk management. NHS stakeholders view ISO 27001 as a **gold standard for protecting sensitive patient data** [19]. In fact, having ISO 27001 can streamline completing the DSP Toolkit, as many controls overlap. For OTOS, investing in ISO 27001 compliance will signal a strong commitment to data security – crucial when dealing with health or probation data that may be sensitive.

- **Digital Technology Assessment Criteria (DTAC):** The **NHS DTAC** is a relatively new (launched 2021) assessment framework that covers five areas: **clinical safety, data protection, technical security, interoperability, and usability/accessibility** [20] [21]. NHS England uses DTAC as a **baseline criteria for digital health tools** – essentially a checklist that products must satisfy to be recommended or procured by NHS organizations [22]. Completing a DTAC assessment means OTOS would need to provide evidence of things like clinical risk management processes (e.g. complying with DCB0129 clinical safety standards), GDPR compliance and privacy-by-design, cybersecurity measures (e.g. encryption, pen-testing), use of open data standards (FHIR, open APIs), and meeting accessibility guidelines (e.g. WCAG for user interfaces). **Having a DTAC "pass" is seen as a NHS stamp of approval** for digital tools entering the system [23]. We strongly advise aligning OTOS with DTAC requirements early – for example, appointing a Clinical Safety Officer and documenting a clinical safety case, preparing a DPIA (Data Protection Impact Assessment), and ensuring the app can integrate with NHS systems. By meeting DTAC, OTOS will satisfy most of the NHS's expectations on safety and tech standards.

- **Other IG and Security Expectations:** In addition to the above, OTOS should ensure compliance with standard UK data protection law (UK GDPR/DPA 2018) and NHS-specific IG policies. The **NHS Information Governance Toolkit** (now DSPT) is supplemented by requirements like confidentiality (Caldicott principles) and records management standards. Moreover, working with the Criminal Justice system (Probation Service) means OTOS must also handle any offender data under MoJ data protection rules and safeguarding obligations. OTOS might also pursue **Cyber Essentials Plus** certification (a UK government-backed cybersecurity standard) as many public sector contracts require it. All these efforts will contribute to an overall posture of trust and security. In summary, **demonstrable compliance is not just best practice but often a prerequisite for NHS partnerships** – e.g. DSP Toolkit compliance is required to sign an NHS data sharing agreement [18], and commissioners increasingly ask for ISO 27001 and DTAC during procurement [24].

Dashboard & Reporting Design for KPIs (Engagement, Outcomes, Safeguarding)

To satisfy stakeholders in NHS England and the Probation Service, OTOS should implement robust **analytics dashboards and reporting capabilities** aligned to key performance indicators (KPIs):

- **User Engagement Metrics:** Measuring engagement is crucial to demonstrate that the target population (patients or probation clients) are actually using the digital therapy. OTOS's dashboard should track metrics such as: **number of users enrolled**, **active usage rates** (e.g. percentage of users logging in weekly), **session counts and duration**, **module completion rates** (for therapeutic modules or exercises), and **drop-off points**. By visualizing engagement trends, OTOS and its NHS/probation partners can identify if users are staying engaged or if there are high attrition rates that need addressing. For example, the tool can report the proportion of users who complete the full programme vs. those who start but disengage after a few days. High engagement is often linked to better outcomes in digital interventions [25], so these metrics will be closely watched. The system should allow drilling down by cohort (e.g. by probation region or NHS clinic) to pinpoint where engagement is strong or weak. **KPIs** here might include "**% of users who complete at least X sessions**" or "**average days of use per user per month**". Presenting these in a clear, graphical format (trend lines, bar charts) will help stakeholders quickly grasp user uptake.

- **Clinical and Rehabilitative Outcomes:** Ultimately, both NHS commissioners and Probation will look for evidence that the digital therapy is achieving its intended outcomes. OTOS should therefore capture and report on **outcome measures** relevant to its intervention. In a health context, this could include **clinical assessments** (for example, improvement in PHQ-9 depression scores or GAD-7 anxiety scores if the tool addresses mental health) measured before and after the intervention. For a probation context, outcomes could extend to **behavioral and social metrics** – e.g. reductions in re-offending rates, improved compliance with probation requirements, or positive lifestyle changes. The dashboard should allow tracking of **outcomes over time** and across participants, ideally showing aggregate improvement and possibly comparisons to those not using the tool. If the tool is part of a formal evaluation, reporting might include **pre/post intervention comparisons** or % of users achieving a defined goal (such as remaining substance-free, if applicable). It's important to link engagement with outcomes in the reports, as stakeholders will want to see that better engagement leads to better results. NHS England is increasingly focusing on evidence of efficacy – for digital tools this ties into NICE's **Evidence Standards Framework**, which defines what level of evidence is needed to show effectiveness and value [26]. Therefore, OTOS should design its data collection to produce the kind of evidence needed (for instance, validated scales, qualitative feedback, etc.), and include those outcomes in regular reports. Clear outcome dashboards (with before/after statistics, response scores, etc.) will help satisfy NHS commissioners that the tool delivers clinical benefit, and satisfy Probation that it aids rehabilitation.

- **Safeguarding and Risk Reporting:** Given the vulnerable nature of many users (especially in probation cohorts or those with mental health needs), **safeguarding is a top priority**. OTOS must have features to detect and act on safeguarding concerns – and these should be reflected in reporting. For example, the platform might include automatic flags if a user reports suicidal ideation, or if their behavior in the app raises concern (e.g. repeatedly indicating high distress). The **safeguarding KPIs** could include: **number of safeguarding alerts generated**, types of issues (suicide risk, safeguarding of others, etc.), and the **response times** or outcomes of those alerts (e.g. "85% of high-risk alerts were reviewed by a clinician within 1 hour"). Probation services will want assurance that any risk of harm to the client or others is being caught and addressed promptly. The dashboard could have a dedicated section for safeguarding incidents, accessible to authorized staff, showing real-time alerts and a log of actions taken (for example, "User X flagged for self-harm risk on [date], escalated to clinical team, outcome: referred to crisis service"). Ensuring **audit trails** for these incidents is also important for compliance. Additionally, from a data perspective, aggregated safeguarding data can inform service improvements (e.g. if certain themes are common, such as domestic abuse concerns, the service can tailor its content or liaise with relevant support services). In reporting to NHS England or HMPPS (Her Majesty's Prison and Probation Service), OTOS should highlight how it contributes to keeping people safe – this might be a qualitative narrative alongside quantitative metrics. Embedding safeguarding into the KPI framework underscores that the tool is not just effective, but also safe and responsible.

- **Reporting Format and Frequency:** We recommend OTOS offers a **dashboard interface for live data** as well as regular summary reports (monthly or quarterly) that can be shared with stakeholders. The dashboard could be web-based, with role-based access (ensuring only appropriate NHS or probation staff see identifiable data, in line with IG). For formal reporting, OTOS might generate PDF or slide-deck summaries of KPIs each quarter, highlighting key stats like engagement levels, outcomes achieved, and any critical incidents or safeguarding actions. These reports should map to the contractual KPIs agreed with NHS England and the Probation Service. For example, if a contract KPI is "75% of users complete the 12-week program", the report should clearly state the completion percentage to date. Visual cues (red/amber/green status) can be used to indicate if targets are on track. Where possible, **benchmarking** can be included – e.g. comparing engagement rates in OTOS to typical benchmarks for digital health interventions – to give context. The reporting should also feed into NHS England's outcomes frameworks and Probation's performance metrics. By structuring the data around the priorities of both healthcare and criminal justice stakeholders, OTOS can demonstrate its value across the board (health improvements and reduced recidivism), thereby securing ongoing support and funding.

In summary, a data-driven approach to **monitoring engagement, outcomes, and safeguarding** will help OTOS meet the expectations of NHS England (which is focused on service effectiveness and patient safety) and the Probation Service (focused on rehabilitation outcomes and risk management). The combination of strong integration with NHS systems, rigorous compliance with data standards (ISO 27001, DSP Toolkit, DTAC), and transparent KPI reporting will position OTOS as a trustworthy and effective digital therapy solution in the eyes of both partners.

Recommendations for OTOS (with DO IT / SHOULD / COULD Classification)

- **DO IT: Integrate with NHS clinical systems via IM1/GP Connect.** Complete the NHS England IM1 pairing process to connect OTOS with GP systems (EMIS and SystmOne) for two-way data sharing [1] [27]. This includes fulfilling all **clinical safety and IG prerequisites** and obtaining partner API licences [5]. This is essential for embedding OTOS into NHS workflows (e.g. updating GPs with patient progress).
- **DO IT: Achieve NHS data security compliance.** Complete the NHS **Data Security & Protection Toolkit** self-assessment (annual requirement) to demonstrate good data handling – this is **mandatory for any NHS supplier** [18]. In parallel, obtain **ISO 27001** certification for information security management (the internationally recognized standard), as it's considered **critical for handling sensitive patient data** [19] and will assure both NHS and justice stakeholders of OTOS's security.
- **DO IT: Complete a DTAC assessment.** Ensure OTOS meets the **Digital Technology Assessment Criteria**, covering clinical safety, data protection, technical security, interoperability and accessibility. A passed DTAC gives a formal "stamp of approval" needed for NHS adoption [23]. Allocate resources to produce evidence (clinical safety case, penetration test results, accessibility compliance, etc.) for this checklist.
- **SHOULD: Use NHS-standard APIs and identity services.** Architect OTOS to use **FHIR-based APIs** for any data exchange (to align with NHS interoperability standards [7]) and integrate **NHS Login** for user authentication. For example, enabling NHS Login allows OTOS to plug into the NHS App ecosystem, as seen with eConsult's in-app integration [10] [11]. This will provide a seamless experience (users can access OTOS via their NHS App with one login) and satisfies NHS expectations for using national services.
- **SHOULD: Implement comprehensive KPI dashboards.** Build an internal **analytics dashboard** tracking: user enrollment, active usage, completion rates, and outcome measures. Ensure it highlights **engagement** (e.g. % of users meeting usage targets), **clinical outcomes** (e.g. improvements in validated scores), and **safeguarding alerts** (frequency and resolution). This dashboard should be accessible to authorized NHS England and Probation officials, providing real-time transparency into performance. Regular summary reports drawn from this data should be prepared for quarterly review meetings.
- **SHOULD: Align outcomes with NHS & Probation targets.** Map the reporting in OTOS to the KPIs required by commissioners. For NHS England, emphasize health outcomes and patient satisfaction; for Probation, include metrics like compliance rates and any indicators of reduced reoffending or improved well-being of participants. Incorporate **NICE's evidence framework** guidance to collect appropriate evidence of effectiveness [26]. By aligning with these targets, OTOS can directly demonstrate how it contributes to broader system goals (e.g. NHS Long Term Plan metrics or HMPPS rehabilitation outcomes).
- **COULD: Enhance safeguarding integration.** As an added measure, integrate OTOS's alert system with external systems: for instance, automatically sharing high-risk alerts with local NHS crisis teams or probation officers (with proper consent and governance). While maintaining privacy, this could involve secure messaging or API integration with relevant care coordination systems. This proactive approach would go beyond basic reporting, actively helping services intervene when risks are identified.
- **COULD: Leverage advanced analytics for insights.** Use the data collected to perform deeper analysis, such as identifying which user engagement patterns predict the best outcomes, or using AI to flag individuals who may disengage or deteriorate. These insights (shared in reports) could help NHS and Probation partners target resources (e.g. who might need additional

support). Though not initially required, developing these analytics could increase OTOS's value proposition.

- **COULD: Pursue broader integrations.** In the longer term, OTOS could integrate with **HMPPS systems** (e.g. probation case management software) to streamline referrals and data sharing between healthcare and criminal justice. It could also connect with NHS England's **population health platforms** to contribute de-identified outcome data for service evaluations. Such integrations would further embed OTOS into the service landscape and open up opportunities for scale.

Deep Research Sync Complete – updates sent to BR7(2).

---

1 2 3 4 9 27 Interface Mechanism 1 API standards - NHS England Digital
https://digital.nhs.uk/developer/api-catalogue/interface-mechanism-1-standards

5 6 IM1 Pairing integration - NHS England Digital
https://digital.nhs.uk/services/digital-services-for-integrated-care/im1-pairing-integration

7 8 FHIR (Fast Healthcare Interoperability Resources) - NHS England Digital
https://digital.nhs.uk/services/fhir-apis

10 11 14 15 Our Integrations - eConsult
https://econsult.net/our-partners/our-integrations

12 13 NHS App partners and services currently integrating - NHS England Digital
https://digital.nhs.uk/services/nhs-app/how-to-integrate-with-the-nhs-app/nhs-app-partners-and-services-currently-integrating

16 17 Data Security and Protection Toolkit - NHS England Digital
https://digital.nhs.uk/services/data-security-and-protection-toolkit

18 19 23 24 Top Digital Health Compliance Standards to Watch in 2025
https://www.periculo.co.uk/cyber-security-blog/top-digital-health-compliance-standards-to-watch-in-2025

20 21 22 26 Digital Technology Assessment Criteria (DTAC) - Key tools and information - NHS Transformation Directorate
https://transform.england.nhs.uk/key-tools-and-info/digital-technology-assessment-criteria-dtac/

25 Engagement and retention in digital mental health interventions
https://bmcdigitalhealth.biomedcentral.com/articles/10.1186/s44247-024-00105-9